
Executable Evaluation Contracts for Target-Weight Trading Pipelines

Kevin Lee 

University of California, Los Angeles
micro1
kevinlee69720@g.ucla.edu

Abstract

A sequence of target weights is not a complete backtest specification. Reported performance also depends on when signals become executable, what residual cash earns, how turnover incurs cost, and whether the comparator bears the same exposure. We encode these choices as executable contracts in QUANTCORTEX, an open-source research and paper-trading system. The contracts cover causal inputs, bounded targets, adjusted-close pseudo-share execution, explicit cash and costs, separate attribution and tradable comparators, fail-closed order state, and provenance for every published result artifact.

We apply the protocol to a fixed six-ETF rotation strategy over 2,011 U.S. trading sessions from 2018–2025. Under the audited conventions, the strategy records a 1.40% net CAGR and a -0.15 conventional sample Sharpe relative to a short-Treasury ETF cash proxy (SHV). A causal target-exposure comparator, after its own modeled costs, records 5.72% and 0.62; the strategy’s annualized arithmetic shortfall is -4.14% [-6.70%, -1.63%]. An exact daily identity decomposes its -0.90% annualized excess over cash into 3.90% from passive risky exposure, -3.38% from active risky allocation, 0.30% from exposure timing, and -1.72% from modeled cost. The active-allocation interval is [-5.85%, -0.90%]. Holding the targets and data fixed, omitting costs changes the Sharpe to 0.14; deliberately assigning a close-derived target to the return ending at that close changes it to 0.03.

This is an evaluation methodology and a negative case study, not a new alpha model. It shows that deterministic, artifact-traced computation can still produce an economically misleading estimate when timing, cash, costs, and comparator exposure are left implicit.

1 Introduction

A target-weight vector is a convenient interface between prediction, portfolio construction, and execution. Qlib and FinRL use modular research pipelines [Yang et al., 2020, Liu et al., 2020]; FinRL-X extends the interface through broker execution [Yang et al., 2026]. A target, however, is an intention rather than a realized return. A complete evaluation must also state when that target can be held, what uninvested capital earns, how turnover becomes cost, which holdings are credited with each return interval, and what happens when an order outcome is uncertain.

These choices correspond to separate hypotheses. The information set may be causal while the cash account is incomplete. The accounting may be correct while the code is nondeterministic. A reproducible backtest may still offer no investment value, and a valid simulation says little about whether retrying an uncertain broker request is safe. Treating these questions as one notion of “correctness” obscures the source of failure.

The distinction is particularly important in financial machine learning, where observations are dependent, effective samples are limited, and repeated strategy search creates substantial false-discovery risk [White, 2000, Harvey et al., 2016, Bailey and López de Prado, 2014, Bailey et al., 2017, Arnott et al., 2019]. We ask a narrower systems question: how can a target-weight pipeline make its economic and operational semantics executable, testable, and traceable from the information set to paper-order state?

The primary contribution is the evaluation methodology. The strategy is a test case, and the software is a reference implementation. Specifically, we (i) define a machine-readable evaluation contract and canonical target tape for information timing, weights, overlays, execution lag, cash, costs, comparison, and uncertain order outcomes; (ii) formalize an exact daily return decomposition for portfolios with changing risky exposure and preserve that identity under joint block resampling; and (iii) bind the fixed case-study evidence to its configuration, source tree, environment, input digest, and published artifacts. The empirical result is intentionally negative: it localizes why one fully traced rotation strategy fails rather than presenting it as profitable alpha.

2 Related work

Financial backtest validity. Sharpe-ratio inference depends on distributional and serial-dependence assumptions [Lo, 2002]. Data snooping, repeated trials, and researcher degrees of freedom can make a selected backtest appear stronger than the process that generated it [White, 2000, Bailey and López de Prado, 2014, Bailey et al., 2017, Harvey et al., 2016]. Arnott et al. [2019] organize these risks into a research protocol. In a recent preprint, Zhang et al. [2026] estimate decision-time leakage by changing one evaluation convention at a time across multiple models and panels. Our same-close diagnostic is narrower: it is one contract test inside a broader accounting and execution audit, not a general leakage benchmark.

Attribution and matched comparison. Performance evaluation has long separated investment decisions into interpretable return components [Fama, 1972, Brinson and Fachler, 1985]. We apply that principle to a portfolio whose risky exposure changes over time. The identity separates passive exposure, exposure timing around the sample mean, active allocation, and modeled cost. It is an exact arithmetic decomposition, not a new geometric linking method. The case-study signal draws on cross-sectional, time-series, and residual momentum ideas [Jegadeesh and Titman, 1993, Moskowitz et al., 2012, Blitz et al., 2011], but is used only as a fixed test vehicle.

Research software and implementation risk. Qlib and FinRL provide modular financial-AI workflows [Yang et al., 2020, Liu et al., 2020]; FinRL-X emphasizes a shared weight interface across research and deployment [Yang et al., 2026]. Computational reproducibility remains difficult even with code and data. Across 1,000 computational reproduction attempts in finance, the original code regenerated the reported result exactly only 52% of the time [Pérignon et al., 2024]. In a recent preprint, Yin et al. [2026] compare five engines across 15 strategies and find material divergence from implementation choices, including transaction costs. Our two-engine result is deliberately more limited: it is an internal parity check for one configuration, not an estimate of industry-wide implementation risk. The manifest and source fingerprints follow the broader reproducibility objectives of Pineau et al. [2021]. Our contribution lies in connecting those artifacts to explicit economic contracts within one executable evaluation. No individual invariant or accounting identity is claimed as new; the contribution is their executable integration and controlled stress test.

3 Executable audit protocol

We use “contract” in the design-by-contract sense: an executable invariant with an explicit failure response [Meyer, 1992]. Figure 1 summarizes the evaluation path. Each stage can fail independently, and a later stage does not repair an invalid earlier one. Table 1 states the checks used in this study.

3.1 Portfolio, execution, and cash contracts

Let $w_d \in \mathbb{R}^N$ be the risky-asset target decided after observing the close at decision time d , and let $e(d)$ be the first available execution bar strictly after d . At $e(d)$, the primary event engine solves

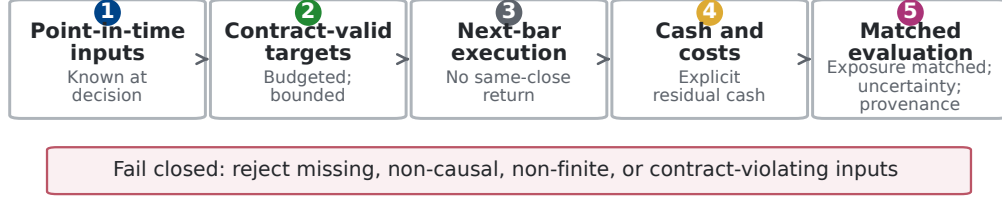


Figure 1: Audit flow from point-in-time inputs to matched evaluation. The audited path validates required inputs, timing, finiteness, and contracts at the stage where each condition becomes testable.

Table 1: Core invariants and failure behavior.

Invariant	Operational check	Failure behavior
Information	Features and fundamentals must be observable by the decision timestamp.	Reject future-dated merges; do not backfill.
Allocation	Allocator output is finite, bounded, and budget-consistent.	Raise a contract error.
Overlay	Timing and risk overlays may reduce exposure but cannot create an invalid book.	Reject non-finite or over-limit weights.
Timing	A close-derived target earns returns only after the first strictly later execution bar.	Shift holdings; never use the observed close twice.
Accounting	Risky assets, residual cash, turnover, and costs share one capital clock.	Reject missing cash bars or cost models.
Comparison	Exact attribution uses realized exposure; a tradable control follows target exposure and pays costs.	Label ex-post controls and cost implementable comparators.
Execution	Persist intent before broker submission; uncertain outcomes are not retried automatically.	Block until broker reconciliation.

adjusted-close pseudo-shares $q_{i,e(d)}$ against post-cost NAV so that the resulting risky allocation matches w_d . It holds q_i fixed between rebalances. If $P_{i,t}$ is the adjusted close and V_t is portfolio NAV, the realized risky weight at the prior close is $h_{i,t-1} = q_{i,t-1}P_{i,t-1}/V_{t-1}$. Close-derived targets therefore earn neither the close they observe nor the return ending at their execution close. For the long-only case, $h_{i,t} \geq 0$ and $\sum_i h_{i,t} \leq 1$; the residual cash holding is

$$h_t^c = 1 - \sum_{i=1}^N h_{i,t}. \quad (1)$$

For simple asset returns $r_{i,t}$ over $(t-1, t]$ and cash-account return r_t^c , pre-cost portfolio return is therefore

$$r_t^{\text{gross}} = \sum_{i=1}^N h_{i,t-1} r_{i,t} + h_{t-1}^c r_t^c. \quad (2)$$

Missing cash observations fail the run instead of silently becoming zero. For executed risky-notional change $\Delta a_{i,t}$ measured against pre-trade NAV, define $B_t = \sum_i \max(\Delta a_{i,t}, 0)$ and $S_t = \sum_i \max(-\Delta a_{i,t}, 0)$. One-way turnover is $\max(B_t, S_t)$, gross traded notional is $B_t + S_t$, and the charge as a fraction of pre-trade NAV is $\kappa_t = 0.0013(B_t + S_t)$. If V_t^- is pre-trade NAV and V_{t-1} is prior-close NAV, the return drag is $d_t = \kappa_t V_t^- / V_{t-1}$ and $r_t^{\text{net}} = r_t^{\text{gross}} - d_t$. Target shares are solved against post-cost NAV. It does not separately estimate commissions, bid-ask spread, taxes, or market impact. This proportional model is a first-order sensitivity assumption; large or urgent orders require models that account for order size, volatility, and execution urgency, such as Almgren and Chriss [2001].

We report nominal CAGR, maximum drawdown, annualized one-way turnover, annualized gross traded notional, and the conventional sample cash-excess Sharpe statistic

$$SR_c = \sqrt{252} \frac{\text{mean}(r_t - r_t^c)}{\text{sd}(r_t - r_t^c)}. \quad (3)$$

Equation 3 is a descriptive statistic, not an iid-normal inference procedure; serial dependence is handled separately through circular-block resampling [Lo, 2002]. An SHV adjusted-close series supplies the return credited to residual cash; the engine does not trade SHV when cash weight changes. It is a short-duration cash proxy, not a frictionless risk-free asset. Because the input contains adjusted closes, the pseudo-shares are total-return accounting units with implicit distribution and corporate-action adjustment, not broker-faithful nominal shares.

3.2 Exact return attribution

Headline performance does not identify which decision created or destroyed value. Let $x_t = \sum_i h_{i,t-1}$ be risky exposure during return period t , let b_t be the return of an equal-initial-weight buy-and-hold basket, and let $\bar{x}$ be the full-sample mean of x_t . Define the daily exposure-matched passive return and the constant-exposure diagnostic as

$$m_t = x_t b_t + (1 - x_t) r_t^c, \quad (4)$$

$$k_t = \bar{x} b_t + (1 - \bar{x}) r_t^c. \quad (5)$$

Then net excess over cash has the exact arithmetic decomposition

$$\begin{aligned} r_t^{\text{net}} - r_t^c &= \underbrace{(r_t^{\text{gross}} - m_t)}_{\text{active risky allocation}} + \underbrace{(m_t - k_t)}_{\text{dynamic exposure timing}} \\ &+ \underbrace{(k_t - r_t^c)}_{\text{passive risky exposure}} + \underbrace{(r_t^{\text{net}} - r_t^{\text{gross}})}_{\text{implementation cost}}. \end{aligned} \quad (6)$$

The first term includes group selection, within-risky-asset weighting, and any return effect of the engine’s between-rebalance weight semantics; it is not pure security selection in the Brinson sense. The timing term measures daily exposure changes relative to the sample-average exposure. Because $\bar{x}$ is estimated over the full evaluation window, k_t is an ex-post diagnostic rather than a tradable ex-ante benchmark. Equation 6 holds on every date, so arithmetic annualized means add exactly. CAGR does not admit the same additive interpretation.

We resample all five series in Equation 6 with common circular-block indices. Consequently, the identity also holds within every bootstrap draw. The reported fraction of positive draws is a resampling diagnostic, not a posterior probability or a multiple-testing-adjusted p -value.

3.3 Single-assumption diagnostics

Three diagnostics change one assumption while holding the targets, primary engine, and input matrix fixed: residual cash earns zero, proportional costs are removed, or a close-derived target is deliberately executed one bar before that close so it earns the return ending at the observed close. The last construction uses future information and is therefore invalid. Under explicit share holdings, changing the cash return can also change realized exposure between rebalances through the NAV denominator. These are sensitivity checks, not candidate strategies, and they play no role in model selection. The paired design resembles the one-switch logic of Zhang et al. [2026], but here the switches audit cash, costs, and execution semantics in one pipeline.

3.4 Reference implementation and order-state controls

The reference implementation separates data, alpha, portfolio, timing, risk, backtest, execution, and strategy layers. Allocator outputs satisfy a strict budget contract. Post-overlay outputs use a relaxed exposure contract because a timing or risk gate may intentionally leave capital in cash. Point-in-time merge guards reject fundamentals that were not yet public, and optional broker, machine-learning, and provider SDKs remain lazy imports so the scientific core can be tested offline.

A central manifest records the clean source revision, Python and package versions, dependency lock, input SHA-256, experiment design, source-tree hashes, and the hash of every published table and

Table 2: Experiment specification recorded in the manifest.

Component	Fixed value
Universe	QQQ, VGT, GLD, TLT, SPY, VIG; QQQ benchmark
Rebalance	First observed session of each calendar week
Group ranking	126-session active-return information ratio; top 2 groups
Residual momentum	126-session estimation + 126-session formation; 21-session gap
Allocation	Positive scores, proportional long-only weights, 60% asset cap
Regime overlay	Rolling vol feature, up to 20 sessions; seeded 3-component full-covariance GMM; 100 EM iterations; 10^{-5} regularization
Volatility overlay	20-point target; 0.3–1.0 multiplier; 21-session proxy
Execution	Event-driven adjusted-close pseudo-shares; first bar strictly after the close-derived decision
Cash and costs	SHV return credited to residual cash; 13 bps of pre-trade gross traded notional
Evaluation	2018-01-02 to 2025-12-31 after 503 warm-up sessions (274 required)
Uncertainty	5,000 joint circular-block draws; seed 42

figure. A versioned JSON contract and canonical long-form target-tape schema make the engine boundary machine readable; deterministic fixtures exercise it without using the empirical price matrix. The empirical sections below exercise the research and backtest contracts. Order-state controls are verified by deterministic tests: paper-trading state is written atomically as a versioned snapshot, intent is persisted before submission, and an uncertain submission is not retried automatically. SDK-conformance tests instantiate the pinned Alpaca and Interactive Brokers request types. They do not test authenticated transport or venue behavior. Appendix E separates exact identities, property tests, counterfactuals, fault injection, SDK conformance, and untested live behavior.

4 Case study and evaluation design

4.1 Fixed audit configuration

The universe contains QQQ, VGT, GLD, TLT, SPY, and VIG, grouped as growth equities (QQQ, VGT), gold and nominal duration (GLD, TLT), and broad and dividend equities (SPY, VIG). On the first session of each week, the strategy ranks groups by a 126-session information ratio relative to QQQ and keeps the top two. Within selected groups, it estimates a one-factor regression on a preceding 126-session window and sums residual returns over a separate 126-session formation window, skipping the latest 21 sessions. Positive scores receive proportional long-only weights, capped at 60% per asset; if no selected member has a positive score, the target is all cash. QQQ is also one of the two growth-group members, so that group’s active-return spread contains one zero benchmark leg. This asymmetry is a fixed design choice.

Two overlays control risky exposure. A seeded three-component Gaussian mixture maps the latest observation to zero, half, or full exposure. Its inputs are QQQ return and rolling realized volatility configured for 20 sessions, represented both as a decimal and in volatility points. The earliest 19 feature rows use the shorter history then available. The two volatility columns are exact rescalings, not independent signals. At each weekly decision, the model is refit on the expanding history available through that close after at least 60 QQQ return observations. Components are ordered by fitted-sample mean return, and only the latest label controls exposure. A separate scaler targets 20 volatility points using a 21-session QQQ realized-volatility proxy because no VIX series is supplied.

The full strategy uses both overlays. Three named variants remove the regime gate, the volatility scaler, or both. Every experiment-facing parameter is recorded in the manifest, and the source-tree digest detects changes to implementation constants. The configuration was frozen for the reported run, but the project was not preregistered and its earlier research trial count is unknown. We therefore report all named variants and make no multiple-testing-adjusted discovery claim.

4.2 Data, comparators, and uncertainty

The owner-supplied matrix is documented as adjusted closes from 2016-01-04 through 2025-12-31, retrieved on 2026-06-16 through yfinance 1.4.1 with `auto_adjust=False`. The yfinance documentation describes the Yahoo Finance API as intended for personal use and directs users to Yahoo’s terms of use.¹ The owner authorizes publication of the derived aggregates, but the raw matrix is not distributed. The audited loader requires complete rows and performs no forward filling. The manifest records the provider, retrieval date, adjustment setting, and input SHA-256:

```
efb384a62157e56a0cd8065abf45c1ed07d90ec26c681e5d54d74fe4cb9c55e1
```

The first 503 sessions are signal history; evaluation covers 2,011 sessions from 2018-01-02 through 2025-12-31.

Comparators share one capital clock. SPY and a six-ETF buy-and-hold basket initialized at equal dollar weights are measured from the close immediately preceding the evaluation window. The exact attribution control multiplies the basket return by the strategy’s realized risky exposure each day and assigns the remainder to SHV. It is ex-post and gross of comparator costs, so it is used to explain return rather than presented as tradable. Gross-to-gross and net-to-gross differences against this control remain useful attribution quantities.

We separately run a causal, costed comparator. At each original strategy decision time it targets the strategy’s declared gross risky exposure in the same equal-initial-weight buy-and-hold basket, executes on the next bar, assigns residual capital to SHV, and pays the same 13-bps proportional cost. Its initial position is established on the pre-evaluation close, outside the reported window. For Equation 6, the basket is also held at the strategy’s full-sample mean exposure. That constant-exposure series is another explicitly ex-post diagnostic.

For dependent daily observations, we jointly resample return-component rows with the circular-block bootstrap of Politis and Romano [1992]. We use 5,000 replications, a 21-session primary block (approximately one trading month), seed 42, and unstudentized two-sided 95% percentile intervals. We repeat the active-return and conventional sample Sharpe calculations with 5- and 63-session blocks to span roughly one week through one quarter. Sharpe is recomputed inside every draw rather than assigned an iid standard error. Intervals quantify sampling variation conditional on this historical path, strategy specification, cash proxy, and block length. They do not establish stationarity or future performance. Fractions of positive draws are descriptive and are not interpreted as significance tests.

5 Results

5.1 Cash accounting changes performance, not the verdict

Table 3 and Figure 2 separate cash and cost effects. Crediting residual cash raises net CAGR from -0.17% under a zero-return cash assumption to 1.40%. That correction is economically material because mean risky exposure is only 30.14% and the strategy is fully in cash on 47.94% of sessions; the exposure path in Figure 2 makes that changing capital allocation explicit. It does not create evidence of alpha. SHV alone compounds at 2.50%. The realized-exposure attribution control compounds at 6.74% before comparator costs; the causal target-exposure comparator compounds at 5.72% after its own modeled costs.

5.2 Allocation and cost explain the shortfall

Equation 6 changes the interpretation of the headline return. Passive risky exposure contributes 3.90% per year; active allocation contributes -3.38%, exposure timing 0.30%, and modeled cost -1.72%. These components sum exactly to -0.90% annualized excess over SHV before rounding.

Table 4 shows that active allocation is the clearest source of underperformance: its 95% interval is [-5.85%, -0.90%], wholly below zero. The timing interval spans zero, so this sample does not isolate a reliable timing effect. Passive exposure is positive, while modeled cost is negative by construction.

¹<https://ranaroussi.github.io/yfinance/>

Table 3: Audited accounting over 2018–2025. Conventional sample Sharpe statistics use SHV except for the explicitly zero-cash row, which uses a zero reference rate.

Series	CAGR	Sharpe	Max drawdown
Strategy, after costs and SHV cash	1.40%	-0.15	-9.83%
Strategy, before costs and SHV cash	3.17%	0.14	-8.60%
SHV cash proxy	2.50%	–	-0.45%
Realized-exposure attribution control, gross	6.74%	0.80	-4.64%
Target-exposure comparator, after costs	5.72%	0.62	-4.94%
Strategy, after costs and zero cash	-0.17%	0.00	-15.15%

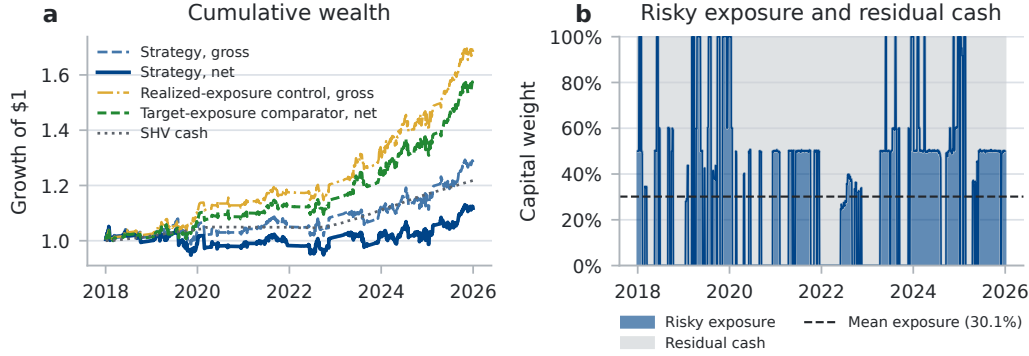


Figure 2: (a) Growth of one dollar under audited cash and cost accounting. (b) The daily capital split between risky assets and residual cash; the dashed line is mean risky exposure. Panel (a) distinguishes the gross ex-post attribution control from the causal target-exposure comparator after costs.

The net-cash interval also spans zero; that does not contradict the stronger matched-comparator result, because the positive passive-exposure component partially offsets allocation and cost losses.

Panel (b) of Figure 3 reports the single-assumption diagnostics. Omitting cash return lowers the CAGR to -0.17%; evaluated relative to SHV, its Sharpe is -0.42. Omitting modeled costs raises the CAGR and Sharpe to 3.17% and 0.14. The deliberately invalid same-close calculation reports 2.49% and 0.03. Thus cost omission and the invalid same-close assignment each change the Sharpe sign relative to the audited value of -0.15. This sign flip is evidence of protocol sensitivity, not evidence that either shortcut is tradable.

5.3 Cost sensitivity and overlay ablations

The strategy turns over 10.30 times per year on a one-way basis and trades 13.25 times NAV per year on a gross two-sided basis. At zero cost, its cash-excess Sharpe is 0.14; it falls to 0.03 at 5 bps, -0.15 at the 13-bps baseline, -0.42 at 25 bps, and -0.96 at 50 bps. The performance estimate is therefore highly sensitive to modest proportional friction.

Appendix Figure 6 compares each variant before costs with a passive basket at the same daily exposure. This avoids attributing the result solely to strategy turnover. The full model has gross cash-excess Sharpe 0.14 versus 0.80 for its comparator. Removing the regime gate, the volatility scaler, or both increases strategy exposure and raw returns, but every gross strategy still trails its matched comparator. The corresponding annualized gross active returns are -3.38%, -5.49%, -3.71%, and -5.54%; every 21-session block-bootstrap interval remains below zero. No variant is selected as a replacement model.

5.4 Uncertainty, subperiods, and engine parity

The net strategy’s annualized arithmetic excess return over SHV is -0.90%, with a 95% block-bootstrap interval of [-4.77%, 3.14%]. Before costs, the corresponding estimate is 0.82% [-3.02%,

Table 4: Exact attribution of annualized arithmetic net excess over SHV. Intervals use common 21-session circular-block draws. “Positive” is the number of resampled annualized means above zero, not a posterior probability.

Component	Mean	95% interval	Positive
Active risky allocation	-3.38%	[-5.85%, -0.90%]	15/5000
Dynamic exposure timing	0.30%	[-2.57%, 3.26%]	2889/5000
Passive risky exposure	3.90%	[0.82%, 6.85%]	4963/5000
Modeled implementation cost	-1.72%	[-2.08%, -1.38%]	0/5000
Net excess over cash	-0.90%	[-4.77%, 3.14%]	1682/5000

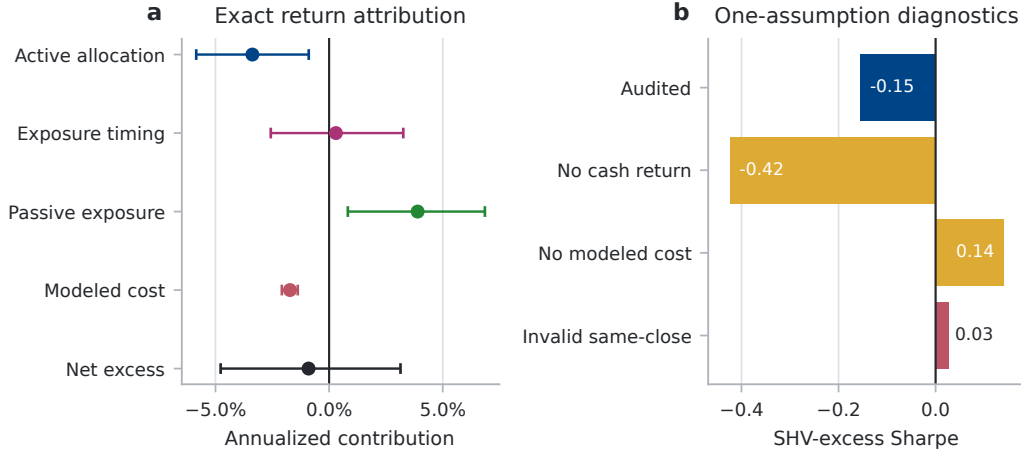


Figure 3: (a) Exact arithmetic return attribution with primary block-bootstrap intervals. (b) One-assumption diagnostics. Amber bars change economic assumptions; the red same-close bar violates causality. None is a strategy alternative.

4.80%]. Neither supports a positive cash-relative claim. Its conventional sample Sharpe is -0.15, with a direct block-bootstrap interval of [-0.80, 0.56].

Relative to the causal comparator after both sides pay modeled costs, annualized arithmetic active return is -4.14% [-6.70%, -1.63%], and conventional active Sharpe is -0.92 [-1.47, -0.37]. Relative to the realized-exposure attribution control, gross active return is -3.38% [-5.85%, -0.90%], and only 0.30% of primary bootstrap draws are positive. After costs it is -5.10% [-7.72%, -2.54%], with no positive primary draw among 5,000.

Appendix Figure 4 shows that changing the circular-block length from 5 to 21 or 63 sessions does not change the matched-comparator conclusion: all before- and after-cost intervals remain below zero.

The costed target-exposure comparator also outperforms in both fixed calendar halves (subperiods.csv). The adjusted-close pseudo-share event engine is primary. Its net CAGR and Sharpe are 1.405% and -0.1549, versus 1.404% and -0.1547 for the vectorized diagnostic. This close agreement is local and does not make their rebalancing semantics equivalent.

6 Limitations and broader impact

6.1 Statistical validity

The configuration was frozen only before the reported run. The project was not preregistered, and no complete log of earlier strategy trials survives. Reporting all four named variants prevents selection within that set but cannot recover unknown prior trials; a Deflated Sharpe Ratio or Reality Check would therefore create false precision [White, 2000, Bailey and López de Prado, 2014]. The block bootstrap conditions on one path and approximate dependence through the chosen block length.

Its positive-draw fractions are descriptive, and its marginal intervals are not multiplicity-adjusted. Stability across 5, 21, and 63 sessions is sensitivity evidence, not proof of stationarity or future performance. Component interval endpoints need not add, although the decomposition holds within every draw. The conventional $\sqrt{252}$ Sharpe remains a descriptive statistic; block resampling addresses uncertainty but does not remove the weak-dependence and stationarity assumptions needed to interpret that uncertainty.

6.2 Economic construct validity

“Active risky allocation” combines group selection, within-risky weighting, and the engine’s between-rebalance semantics. “Dynamic exposure timing” is relative to full-sample mean exposure, hence diagnostic rather than tradable. The realized-exposure attribution control is not tradable and excludes its own costs. The target-exposure comparator is causal and costed, but its risky leg is a constructed buy-and-hold basket rather than a quoted security; its cost model still omits spread and impact. Arithmetic means decompose exactly; compounded wealth does not.

SHV is only a cash proxy: it has duration, expenses, and market-price variation. Adjusted-close pseudo-shares embed total-return adjustments and omit broker share records, spreads, queue position, and fill uncertainty. The proportional cost grid has no order-size, volatility, venue, impact, ADV, or capacity term. Without VIX, the regime model receives realized volatility twice in different units; its covariance regularization is scale-sensitive, and the separate volatility scaler uses the same underlying proxy.

6.3 External validity and reproducibility

The study covers one U.S.-listed ETF universe, one strategy family, and one eight-year path; it does not estimate failure magnitudes for other settings or match broader multi-model and multi-engine studies [Zhang et al., 2026, Yin et al., 2026]. The provider can revise adjusted histories. The raw input is not redistributed, and provider permission for the derived publication has not been independently verified. Open schemas and conformance fixtures reproduce software semantics, not these returns; an open performance evaluation requires a permitted, reviewer-accessible panel.

6.4 Operational scope and broader impact

Fault-injection tests cover stale or uncertain execution state, but no live authentication, venue behavior, funded order, capacity, or production control was tested. The intended benefit is reduced false confidence; the principal risk is mistaking an audited simulation for investment advice or deployment approval. Paper-mode defaults and fail-closed retries mitigate but do not remove that risk. Past simulated performance does not imply future returns.

7 Conclusion

A target-weight generator does not determine a realized return; information timing, execution, cash, costs, and comparator exposure complete the experiment. Here, crediting residual cash corrects performance but does not create alpha: passive exposure contributes positively, while active allocation and modeled cost erase that benefit. A causal comparator using the same target exposure and cost model reaches the same negative verdict as the exact ex-post attribution. Cost omission or same-close return assignment changes the Sharpe sign without changing the target generator. The contribution is a localized, testable failure analysis rather than a profitable signal.

References

- Robert Almgren and Neil Chriss. Optimal execution of portfolio transactions. *The Journal of Risk*, 3(2):5–39, 2001. doi: 10.21314/JOR.2001.041.
- Rob Arnott, Campbell R. Harvey, and Harry Markowitz. A backtesting protocol in the era of machine learning. *The Journal of Financial Data Science*, 1(1):64–74, 2019. doi: 10.3905/jfds.2019.1.064.

- David H. Bailey and Marcos López de Prado. The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management*, 40(5):94–107, 2014. doi: 10.3905/jpm.2014.40.5.094.
- David H. Bailey, Jonathan M. Borwein, Marcos López de Prado, and Qiji Jim Zhu. The probability of backtest overfitting. *The Journal of Computational Finance*, 20(4):39–69, 2017. doi: 10.21314/JCF.2016.322.
- David Blitz, Joop Huij, and Martin Martens. Residual momentum. *Journal of Empirical Finance*, 18(3):506–521, 2011. doi: 10.1016/j.jempfin.2011.01.003.
- Gary P. Brinson and Nimrod Fachler. Measuring non-U.S. equity portfolio performance. *The Journal of Portfolio Management*, 11(3):73–76, 1985. doi: 10.3905/jpm.1985.409005.
- Eugene F. Fama. Components of investment performance. *The Journal of Finance*, 27(3):551–567, 1972. doi: 10.1111/j.1540-6261.1972.tb00984.x.
- Campbell R. Harvey, Yan Liu, and Heqing Zhu. . . . and the cross-section of expected returns. *The Review of Financial Studies*, 29(1):5–68, 2016. doi: 10.1093/rfs/hhv059.
- Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1):65–91, 1993. doi: 10.1111/j.1540-6261.1993.tb04702.x.
- Xiao-Yang Liu, Hongyang Yang, Qian Chen, Runjia Zhang, Liuqing Yang, Bowen Xiao, and Christina Dan Wang. FinRL: A deep reinforcement learning library for automated stock trading in quantitative finance. *arXiv preprint arXiv:2011.09607*, 2020. URL <https://arxiv.org/abs/2011.09607>.
- Andrew W. Lo. The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4):36–52, 2002. doi: 10.2469/faj.v58.n4.2453.
- Bertrand Meyer. Applying “design by contract”. *Computer*, 25(10):40–51, 1992. doi: 10.1109/2.161279.
- Tobias J. Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen. Time series momentum. *Journal of Financial Economics*, 104(2):228–250, 2012. doi: 10.1016/j.jfineco.2011.11.003.
- Christophe Pérignon, Olivier Akmansoy, Christophe Hurlin, Anna Dreber, Felix Holzmeister, Jürgen Huber, Magnus Johannesson, Michael Kirchler, Albert J. Menkveld, Michael Razen, and Utz Weitzel. Computational reproducibility in finance: Evidence from 1,000 tests. *The Review of Financial Studies*, 37(11):3558–3593, 2024. doi: 10.1093/rfs/hhae029.
- Joelle Pineau, Philippe Vincent-Lamarre, Koustuv Sinha, Vincent Larivière, Alina Beygelzimer, Florence d’Alché-Buc, Emily Fox, and Hugo Larochelle. Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program. *Journal of Machine Learning Research*, 22(164):1–20, 2021. URL <https://jmlr.org/papers/v22/20-303.html>.
- Dimitris N. Politis and Joseph P. Romano. A circular block-resampling procedure for stationary data. In Raoul LePage and Lynne Billard, editors, *Exploring the Limits of Bootstrap*, pages 263–270. John Wiley & Sons, New York, 1992.
- Halbert White. A reality check for data snooping. *Econometrica*, 68(5):1097–1126, 2000. doi: 10.1111/1468-0262.00152.
- Hongyang Yang, Boyu Zhang, Yang She, Xinyu Liao, and Xiaoli Zhang. FinRL-X: An AI-native modular infrastructure for quantitative trading. *arXiv preprint arXiv:2603.21330*, 2026. URL <https://arxiv.org/abs/2603.21330>.
- Xiao Yang, Weiqing Liu, Dong Zhou, Jiang Bian, and Tie-Yan Liu. Qlib: An AI-oriented quantitative investment platform. *arXiv preprint arXiv:2009.11189*, 2020. URL <https://arxiv.org/abs/2009.11189>.

Dong Yin, Takeshi Miki, Vladislav Lesnichenko, and Vasyl Gural. Implementation risk in portfolio backtesting: A previously unquantified source of error. *arXiv preprint arXiv:2603.20319*, 2026. URL <https://arxiv.org/abs/2603.20319>.

Fan Zhang, Zhen Li, Sijia Peng, and Yu Chen. When alpha disappears: A one-switch benchmark for decision-time leakage in financial backtests. *arXiv preprint arXiv:2605.23959*, 2026. URL <https://arxiv.org/abs/2605.23959>.

A Reproducibility details

The complete generator is `scripts/run_paper_experiments.py`. The release wrapper checks out the committed source revision in a detached clean worktree, writes experiment outputs outside that worktree, verifies the recorded revision and start-state cleanliness, then builds the paper with Tectonic 0.16.9. The reviewed invocation is:

```
scripts/release_paper_artifacts.sh \
  local_data/published_rotation_prices.csv
```

The input must contain adjusted-close columns QQQ, VGT, GLD, TLT, SPY, VIG, and SHV and match the input digest reported in Section 4 for exact reproduction. Aggregate CSVs, generated LaTeX values, figures, and their hashes are listed in `paper/results/manifest.json`. The manifest also records the clean source commit, dependency lock, canonical configuration hash, thread libraries, contract hash, and every package Python file individually. The aggregate source-tree digest is:

df200735ec35013b252a3aa689752b9eae485c21abe7247bc590c444f18221b6

The public code is available at <https://github.com/magnaquant/quantcortex>. The reviewed run used Python 3.14.4 on an eight-core Apple M1 with 8 GiB RAM and completed in under 40 seconds on CPU; no GPU was used. Supported CI runs use Python 3.11 through 3.14. The reported experiment consumed less than one CPU-minute; repository-wide testing, packaging, notebook execution, and broker conformance checks were additional engineering validation rather than model search.

B Block-length sensitivity

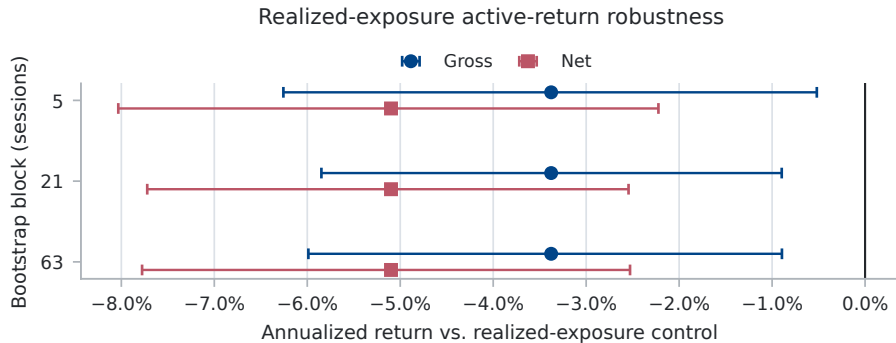


Figure 4: Active-return estimates and 95% percentile intervals under 5-, 21-, and 63-session circular-block resampling against the realized-exposure attribution control. Gross compares before-cost strategy returns with the gross control; net retains modeled strategy costs while the control remains gross.

C Additional engine comparison

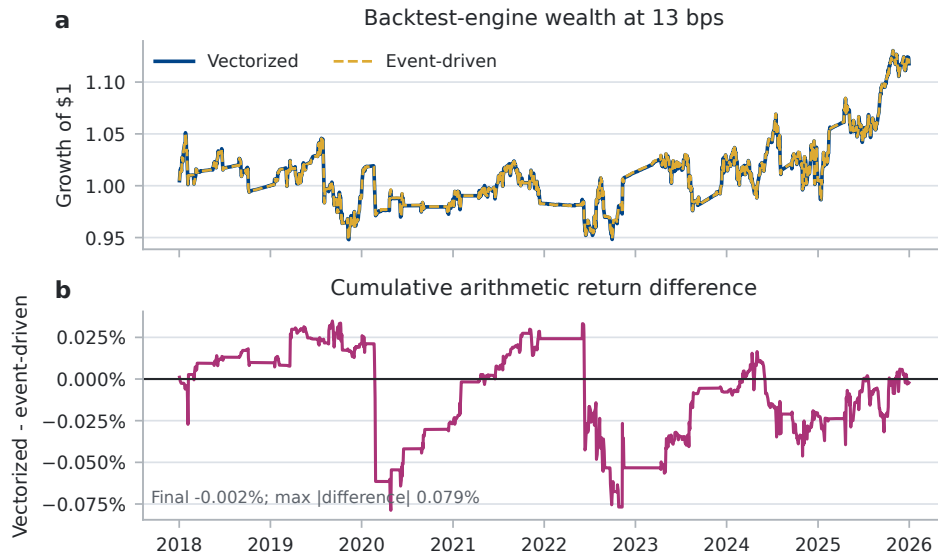


Figure 5: The vectorized engine re-pegs target weights between rebalances, whereas the event-driven engine holds explicit drifting adjusted-close pseudo-shares. Their small difference here does not remove the semantic distinction.

D Cost and overlay sensitivity

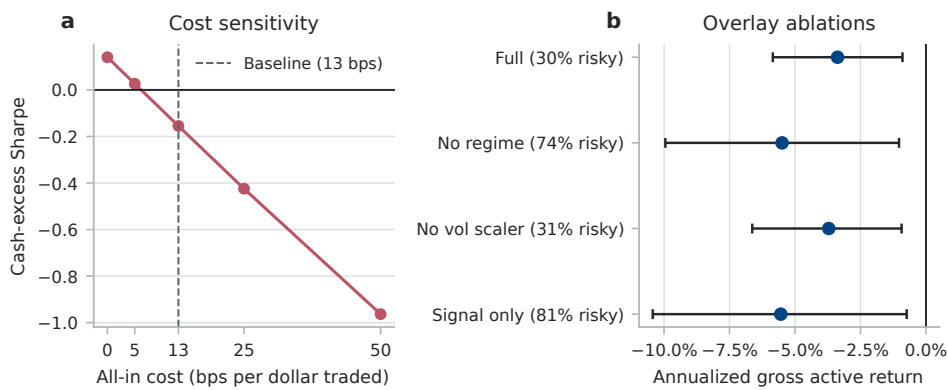


Figure 6: (a) After-cost Sharpe sensitivity for the full strategy. (b) Annualized gross strategy return minus the realized-exposure attribution control, with 95% intervals from common 21-session circular-block draws. Tick labels report mean risky exposure; all named audit variants are shown.

E Contract-to-evidence map

The artifact distinguishes evidence types rather than treating every passing check as equivalent:

Evidence class	Claim supported	Committed evidence
Exact identity	Daily return components reconstruct net cash excess	Per-row assertion and <code>return_decomposition.csv</code>
Property test	Weight, timing, cash, and target-tape invariants hold	Deterministic pytest cases, including hand-computable tapes
Counterfactual	One declared assumption changes the reported metric	<code>protocol_switches.csv</code>
Fault injection	Unsafe or stale execution state fails closed	Truncated reconciliation, unknown status, and stale-writer tests
SDK conformance	Offline request and response mappings match pinned SDKs	Broker model-construction tests and behavioral mocks
Untested live	Authenticated transport, venue state, and funded orders	No supporting evidence; explicitly outside scope

F Use of language-model assistance

The author used an LLM-based coding assistant for repository inspection, code review, regression-test drafting, and copyediting. The author retained responsibility for the strategy, data, statistical methods, experiment design, and all reported claims, and reviewed the resulting code and prose. Numerical claims are generated by committed scripts and machine-readable artifacts.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and Introduction identify the primary contribution as an evaluation methodology and describe the strategy as a fixed negative case study rather than a new alpha model, general benchmark, or deployment claim. All reported values are generated from the experiment artifacts used in Sections 4 and 5.

Guidelines:

- The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Section 6 discusses the single-universe design, attribution and engine semantics, the ex-post attribution control and costed comparator, cash proxy, adjusted-close data, proportional costs, scale-sensitive regime features, bootstrap assumptions, unknown prior trials, data availability, scaling limits, and untested live transport.

Guidelines:

- The answer [\[N/A\]](#) means that the paper has no limitation while the answer [\[No\]](#) means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.

- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper contains accounting definitions and empirical estimators, but no theorem, lemma, or new theoretical result requiring a proof.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [No]

Justification: Sections 3–4 and Appendix A disclose the algorithms, dates, symbols, parameters, versions, clean-worktree release command, and exact input digest. The raw price matrix is not redistributed, however, so independent bitwise reproduction is not fully open.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.

- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The MIT-licensed code, aggregate results, figures, and manifests are open, but the provider-derived adjusted-close matrix is not. Appendix A explains the exact command and required columns for an authorized local copy.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [\[Yes\]](#)

Justification: Section 4 specifies the fixed audit configuration, lookbacks, execution timing, variants, cost grid, warm-up and evaluation windows, separate attribution and costed comparators, bootstrap settings, and the unknown earlier research trial count.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [\[Yes\]](#)

Justification: Sections 3–5 report joint circular-block intervals for arithmetic effects and directly resampled conventional sample Sharpe statistics, including sensitivity to 5-, 21-, and 63-session blocks. The paper states the 5,000 replications, seed, percentile construction, dependence rationale, and descriptive interpretation of positive-draw counts.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not include experiments.
- The authors should answer [\[Yes\]](#) if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [\[Yes\]](#)

Justification: Appendix A reports CPU model, core count, memory, Python version, runtime, GPU use, and distinguishes experiment compute from broader engineering validation.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not include experiments.

- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [No]

Justification: The work uses no human subjects or private personal data and places no live orders. However, as disclosed in Section 4 and item 12, the project has not independently verified that the provider's terms permit public publication of derived aggregates, so it cannot certify conformity in every respect.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 6 discusses the benefit of reducing false confidence and the risk of treating simulated results as investment advice or deployment approval, together with practical mitigations.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper releases no pretrained model, scraped media corpus, personal data, or other high-risk dual-use asset requiring gated access.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: Section 4 identifies Yahoo Finance via yfinance 1.4.1, links the provider usage notice, records the owner's authorization to publish derived aggregates, and states that raw quotes are not redistributed. However, the project has not independently verified provider authorization for public publication, so it cannot certify full terms compliance.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The repository documents the MIT-licensed code, target-tape and evaluation-contract schemas, conformance fixtures, architecture, tests, security policy, release command, aggregate result tables, figures, provenance manifest, and known limitations.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.

- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The research includes no crowdsourcing and no human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The research includes no intervention or data collection involving human subjects, so IRB review is not applicable.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs are not an important, original, or non-standard component of the strategy or empirical method. The use-of-language-model-assistance appendix voluntarily discloses assistance with code review, regression-test drafting, and copyediting.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.